

Stochastic gradient descent implementation for the minimum information dependence model

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1 Minimum information dependence model

The minimum information dependence model is a joint model for a mixed-domain data proposed by [5]: Let $(\mathcal{X}_i, \mathcal{F}(\mathcal{X}_i), dx_i)$ for $i = 1, \dots, d$ be a measure space and denote their product space by $\mathcal{X} = \prod_{i=1}^d \mathcal{X}_i$ and $dx = \prod_{i=1}^d dx_i$. For index i , use the notation $-i$ to indicate the removal of the i -th coordinate, e.g., $x_{-i} = (x_j)_{j \neq i}$, $\mathcal{X}_{-i} = \prod_{j \neq i} \mathcal{X}_j$, and $dx_{-i} = \prod_{j \neq i} dx_j$.

A statistical model (1) together with the constraints (2) and (3) is called *minimum information dependence model*:

$$p(x; \theta, \nu) = \exp \left(\theta^\top h(x) - \sum_{i=1}^d a_i(x_i; \theta, \nu) - \psi(\theta, \nu) \right) \prod_{i=1}^d r_i(x_i; \nu), \quad (1)$$

where $\theta \in \mathbb{R}^K$ is a K -dimensional parameter representing the dependence, $h : \mathcal{X} \rightarrow \mathbb{R}^K$ is a given function, let $r_1(x_1; \nu), \dots, r_d(x_d; \nu)$ be statistical models of marginal densities on $\mathcal{X}_1, \dots, \mathcal{X}_d$ with ν denotes parameters characterizing the marginal densities, and the functions $a_i(x_i; \theta, \nu)$ and $\psi(\theta, \nu)$ are simultaneously determined by constraints

$$\int p(x; \theta, \nu) dx_{-i} = r_i(x_i; \nu), \quad i = 1, \dots, d, \text{ and} \quad (2)$$

$$\int \sum_{i=1}^d a_i(x_i; \theta, \nu) p(x; \theta, \nu) dx = 0. \quad (3)$$

Note that the density (1) is reduced to the independent model $\prod_{i=1}^d r_i(x_i; \nu)$ if $\theta = 0$.

Two important properties of the model are as follows:

1. The model admits various types of the domains of variables (say, continuous / categorical / manifold / etc)
2. The model admits various types of dependence (higher-order interaction / negative interaction / etc...).

1.1 Pseudo likelihood estimation

Despite the complexity of a_i s and $\psi(\theta, \nu)$, there are several methods to estimate the dependence parameter θ . An option is the Besag pseudo likelihood estimate (PLE); see [5] for another option called conditional likelihood estimate and [6] for the semi-parametric likelihood estimate that specializes bi-variate cases. Given the observation $\{x(1), \dots, x(n) : x(u) \in \mathbb{R}^K, \quad u = 1, 2, \dots, n\}$, PLE is an estimate that maximizes

$$\mathcal{L}(\theta) := \sum_{1 \leq u < v \leq n} \sum_{i=1}^d \mathcal{L}_{uv}^i(\theta),$$

where $\mathcal{L}_{st}^i$ is defined by

$$\mathcal{L}_{uv}^i(\theta) := \log \frac{1}{1 + \exp(H_{uv}^i(\theta))}, \quad (4)$$

$$H_{uv}^i(\theta) := \theta^\top h_{uv}^i, \quad (5)$$

$$h_{uv}^i := \{-h(x^{(i)}(u)) - h(x^{(i)}(v)) + h(x(u)) + h(x(v))\}. \quad (6)$$

The statistical properties of PLE are discussed in [5].

2 Stochastic gradient descent

One drawback of the (full-batch) PLE is that it requires computing all $dn(n-1)/2 \mathcal{L}_{uv}^i$ values at each step. This significantly limits the applicability of the minimum information dependence model.

Stochastic gradient descent (SGD) and its variants broaden the applicability of the minimum information dependence model by using a stochastic approximation of the gradient of $\mathcal{L}(\theta)$. For example, even with a sample size of $n = 5 \times 10^6$, the total computation time including 100 bootstrap replications was approximately 6 minutes in Google Colaboratory, whereas the full-batch PLE failed to run in a standard computing environment! See Section 4 for a detailed comparison between the full-batch and SGD-based PLE methods.

The (vanilla) stochastic gradient descent (SGD) [3] is defined as follows: for the t -th step, the SGD update with step size $\eta_t > 0$ is given by

$$\hat{\theta}^{(t)} = \hat{\theta}^{(t-1)} - \eta_t \nabla_{\theta} \mathcal{L}_{U^{(t)} V^{(t)}}^{I^{(t)}}(\hat{\theta}^{(t)}),$$

where $I^{(t)}$ is drawn from the uniform distribution on $\{1, 2, \dots, K\}$, and $(U^{(t)}, V^{(t)})$ is drawn from the uniform distribution on $\{(u, v) : 1 \leq u < v \leq n\}$.

The mini-batch stochastic gradient descent reduces the variance of the optimization path of SGD by using multiple samples at each step: for the t -th step, the mini-batch SGD update with step size $\eta_t > 0$ is given by

$$\hat{\theta}^{(t)} = \hat{\theta}^{(t-1)} - \eta_t \left(\frac{1}{B} \sum_{b=1}^B \nabla_{\theta} \mathcal{L}_{U^{(t,b)} V^{(t,b)}}^{I^{(t)}}(\hat{\theta}^{(t)}) \right)$$

where $I^{(t)}$ is drawn from the uniform distribution on $\{1, 2, \dots, K\}$, and $(U^{(t,b)}, V^{(t,b)})$ are B independent samples drawn from the uniform distribution on $\{(u, v) : 1 \leq u < v \leq n\}$.

The choice of a step size sequence $\{\eta_t : t = 1, 2, \dots, M\}$ has a significant impact on the resulting estimate. Theoretically, for the convergence of stochastic gradient descent, the step size sequence is required to satisfy the decay conditions [3]

$$\sum_{t=1}^{\infty} \eta_t = \infty \quad \text{and} \quad \sum_{t=1}^{\infty} \eta_t^2 < \infty. \quad (7)$$

In practice, a larger step size is preferred in the early stages of optimization to accelerate convergence, whereas a smaller step size is used in the later stages to ensure stability. Therefore, choosing an appropriate step size schedule can be challenging in practice.

2.1 Adaptive moment estimation (Adam)

One of the practically powerful methods to control the step size sequence is Adaptive moment estimation (Adam) [1]. Adam updates the parameter value as follows: For each m , the Adam update with step size $\eta > 0$, decay rates β_1 and β_2 for the moment estimates, and offset ε is given by

$$\begin{aligned} g^{(t)} &= \nabla_{\hat{\theta}^{(t-1)}} \mathcal{L}_{U^{(t)}V^{(t)}}^{I^{(t)}}, \\ m^{(t)} &= \beta_1 m^{(t-1)} + (1 - \beta_1) g^{(t)} \quad \text{and} \quad v^{(t)} = \beta_2 v^{(t-1)} + (1 - \beta_2) \|g^{(t)}\|_2^2, \\ \tilde{m}^{(t)} &= \frac{m^{(t)}}{1 - \beta_1^t} \quad \text{and} \quad \tilde{v}^{(t)} = \frac{v^{(t)}}{1 - \beta_2^t}, \\ \hat{\theta}^{(t)} &= \hat{\theta}^{(t-1)} - \eta \frac{\tilde{m}^{(t)}}{\sqrt{\tilde{v}^{(t)}} + \varepsilon}, \end{aligned}$$

where $I^{(t)}$ is drawn from the uniform distribution on $\{1, 2, \dots, K\}$, and $(U^{(t)}, V^{(t)})$ is drawn from the uniform distribution on $\{(u, v) : 1 \leq u < v \leq n\}$. Surprisingly, the impact of the hyperparameters $(\beta_1, \beta_2, \varepsilon)$ on optimization performance is empirically small. Therefore, these values are typically fixed to their default settings: $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\varepsilon = 10^{-8}$.

2.2 The Ruppert–Polyak–Juditsky averaging

To gain the stability, the average along with the optimization path is used [4, 2]. This is known as the Ruppert–Polyak–Juditsky (RPJ) averaging scheme: The resulting estimate with the burn-in period T_{burnin} is given by

$$\hat{\theta}_{\text{RPJ}}^{(T)} = \frac{1}{T - T_{\text{burnin}}} \sum_{t=T_{\text{burnin}}}^T \hat{\theta}^{(t)}.$$

As the vanilla average is memory-inefficient, we employ the recursive identity in the implementation

$$\hat{\theta}_{\text{RPJ}}^{(t)} = \hat{\theta}_{\text{RPJ}}^{(t-1)} + \frac{1}{t} \left\{ \hat{\theta}^{(t)} - \hat{\theta}_{\text{RPJ}}^{(t-1)} \right\},$$

since we have, for a sequence $\{a_t : t = 1, 2, \dots, T\}$,

$$\frac{1}{T} \sum_{t=1}^T a_t = \frac{1}{T-1} \sum_{t=1}^{T-1} a_t + \frac{1}{T} \left\{ a_T - \frac{1}{T-1} \sum_{t=1}^{T-1} a_t \right\}.$$

Our practical recommendation is a two-step procedure: first, use the Adam optimizer for warm-starting, and then apply the RPJ-averaging SGD with a constant step size.

3 Uncertainty quantification

For the full-batch PLE, there are two options for quantifying estimation uncertainty: the bootstrap and the asymptotic covariance evaluation. The bootstrap simply involves re-running the estimation procedure using resampled versions of the original data. The asymptotic covariance of the full-batch PLE is given by the sandwich estimator

$$\frac{4}{n} \hat{J}_{\text{PLE}}^{-1} \hat{I}_{\text{PLE}} \hat{J}_{\text{PLE}}^{-1},$$

where

$$\begin{aligned} \hat{I}_{\text{PLE}} &= \frac{1}{n} \sum_{u=1}^n \left(\frac{1}{n} \sum_{\substack{v=1 \\ v \neq u}}^n \sum_{i=1}^d \frac{h_{uv}^i}{1 + \exp(\hat{\theta}_{\text{PLE}}^\top h_{uv}^i)} \right) \left(\frac{1}{n} \sum_{\substack{v=1 \\ v \neq u}}^n \sum_{i=1}^d \frac{(h_{uv}^i)^\top}{1 + \exp(\hat{\theta}_{\text{PLE}}^\top h_{uv}^i)} \right), \\ \hat{J}_{\text{PLE}} &= \frac{2}{n(n-1)} \sum_{1 \leq u < v \leq n} \sum_{i=1}^d \frac{h_{uv}^i (h_{uv}^i)^\top}{\{1 + \exp(\hat{\theta}_{\text{PLE}}^\top h_{uv}^i)\}^2}. \end{aligned}$$

See [5] for the derivation.

For the SGD-PLE, the most reliable method for uncertainty quantification is the empirical bootstrap. To reduce the computation time, the parallel computation is employed in the implementation. Asymptotic covariance evaluation requires additional consideration of two sources of randomness: the variability arising from the data itself and the stochasticity introduced by the SGD algorithm. By leveraging the central limit theorem of the Ruppert–Polyak–Juditsky (RPJ) averaging SGD [2], we consider the following surrogate for the asymptotic covariance of $\hat{\theta}_{\text{RPJ}}^T$:

$$\hat{\Sigma}_{\text{RPJ}} := 4 \left(\frac{1}{n} + \frac{1}{T - T_{\text{burnin}}} \right) \bar{J}^{-1} \bar{I} \bar{J}^{-1},$$

where $\bar{J}$ and $\bar{I}$ are unbiased estimates of $\hat{J}_{\text{PLE}}$ and $\hat{I}_{\text{PLE}}$, respectively.

4 Comparison with the full-batch and the SGD PLEs

Here, we compare the full-batch PLE with the SGD-PLE. We use the mixed-variable example in [5] and report the computation time, the estimated parameter values, and the variance estimates (uncertainty quantification; UQ). The marginal distributions are specified as Beta(10, 10) and Poisson(3), respectively. The canonical statistic is

$$h(x_1, x_2) = \frac{x_1}{1 + x_2},$$

and the true parameter value is set to $\theta = -10$. The sample size is $n = 1000$. All computations are performed on Google Colaboratory.

	Full-batch	Adam	Adam + RPJ
Computation time	58.08 s	<u>1.49 s</u>	2.52 s
Computation time (with UQ)	444.33 s	<u>16.11 s</u>	17.89 s
Estimate of θ ($\theta = -10$)	<u>-10.70</u>	-8.80	-11.95
Asymptotic variance (UQ)	1.37	0.39	1.12
Bootstrap variance (UQ)	1.44	2.46	1.41

Table 1: Comparison of computation times, parameter estimates, and uncertainty quantification (UQ) for the full-batch PLE and the SGD-PLE methods.

Table 1 presents the estimation results. The computation time of SGD-PLE is remarkably fast compared with that of the full-batch PLE; approximately 25 to 30 times faster!! The estimation and uncertainty quantification results are also quite similar when using Adam + RPJ, which is one of the reasons we recommend the two-step procedure.

References

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